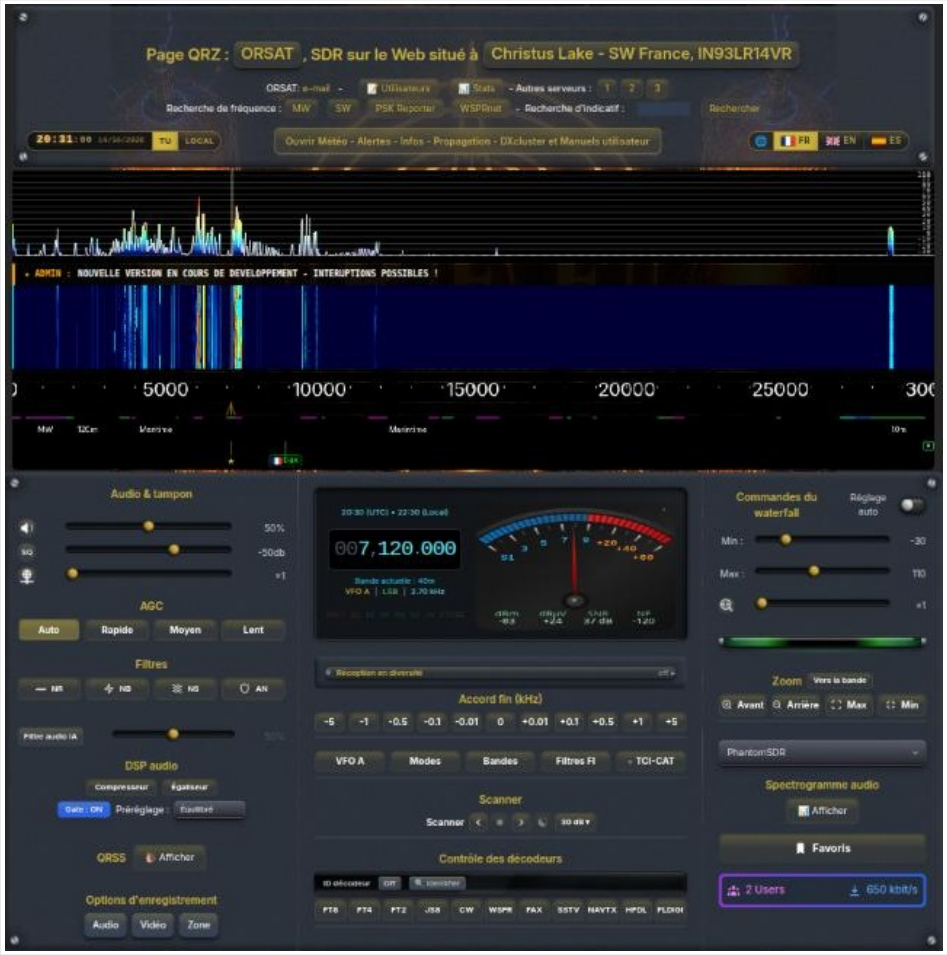
PhantomSDR-Plus “ORSAT”

User Manual – Web Interface

http://orsat.ddns.net:8080



The ORSAT interface as it appears on a desktop screen.

Complete guide to the online radio receiving interface: bands, modes, filters, audio processing, digital decoders, signal identification, receive diversity, recording, bookmarks, DX cluster and companion tools.

English version — October 2026 edition — PhantomSDR-Plus v5.1.0

This version is the result of the collaborative work of **SV1BTL** and **F1NSK**.

The screenshots show the interface in French; switch the interface to English with the EN button (section 1.2).

Contents

1. General introduction 4
1.1 Screen layout 4
1.2 Choosing the language 4
1.3 Tooltips 4
2. Top banner 5
2.1 Station identification line 5
2.2 Second line: contact, users and statistics 5
2.3 Third line: frequency and callsign search 6
2.4 UTC / Local clock 6
2.5 Language and “Information” menu 6
3. Main screen: spectrum and waterfall 7
3.1 Tuning with the mouse and wheel 7
3.2 Listener labels and administrator message 7
4. Dial, bands, modes and filters 8
4.0 Frequency dial, S-meter and indicators 8
4.1 Band selector 9
4.2 Demodulation modes 10
4.3 IF filters 10
4.4 VFO A / B 11
4.5 TCI-CAT 11
4.6 Fine tuning 11
5. Audio control column (left) 12
5.3 Noise filters and AI audio filter 12
5.4 Audio DSP: compressor, equalizer and gate 12
6. QRSS 14
7. Recording 15
8. Receive diversity 16
9. Scanner and decoders 16
9.1 Scanner 17
9.2 Digital decoders 17
9.3 Decoder ID: automatic mode identification 18
9.4 Identify a signal (Artemis database) 19
9.5 FLDIGI decoder (FSK / RTTY and other modes) 20
9.6 HFDL decoder 20
10. Waterfall controls 21

11. Audio spectrogram 21

12. Bookmarks 22

13. “Information” menu 23

13.1 hamdash tools 23

13.2 Setup & configuration 24

13.3 Receiver, antenna and links 24

13.4 User manuals 24

14. DX Cluster 25

15. Tips, troubleshooting and good practice 26

15.1 Listening quality 26

15.2 Audio dropouts or latency 26

15.3 Decoders 26

15.4 Shared use 26

1. General introduction

PhantomSDR-Plus is a software-defined radio (SDR) receiver that you use from an ordinary web browser, with nothing to install. The interface shows the receiver’s RF spectrum and waterfall in real time, and lets you listen to, tune and decode received signals straight from the browser. PhantomSDR aims to be a modern alternative to the PA3FWM WebSDR.

The **ORSAT** station runs PhantomSDR-Plus version 5.1.0 (original project by SV1BTL), modified by F1NSK: “old gold” theme, interface fully translated into French, English and Spanish, DX cluster synchronisation, extra decoders (HFDL, the FLDIGI family), automatic signal identification, receive diversity and built-in hamdash tools.

This manual describes, area by area, every control visible on screen: where it is, what it does and how to use it. Each complex feature comes with a screenshot of the window or panel exactly as it appears in the interface.

1.1 Screen layout

- **Top banner** — station identification, links, clock, language and the “Information” menu (section 2).
- **Central area** — spectrum, waterfall, frequency scale and band markers (section 3).
- **Left column** — audio, AGC, noise filters, DSP, QRSS and recording (sections 5 to 7).
- **Middle column** — frequency dial, S-meter, diversity, fine tuning, bands/modes/filters, scanner and decoders (sections 4, 8 and 9).
- **Right column** — waterfall settings, audio spectrogram, bookmarks and connection indicator (sections 10 to 12).

1.2 Choosing the language

The flag buttons **FR**, **EN** and **ES**, on the right of the top banner (globe icon), instantly switch the whole interface — labels and tooltips — to French, English or Spanish. Your choice is remembered by the browser for future visits.

1.3 Tooltips

Almost every button, slider and area shows an explanatory **tooltip** when you rest the mouse on it for a moment. It is the quickest way to discover a control without leaving the receiving screen.

> ***Tip:*** *The interface is fully reactive: most settings (band, mode, filters, gain) take effect immediately, with no confirmation. Every user has their own tuning and settings, which do not affect the other users connected to the same receiver.*

> ***Note:*** *When the page opens, browsers block sound until the user interacts with it: a “Tap to enable audio” overlay is shown — a single click starts listening.*

2. Top banner

The banner at the very top of the screen identifies the station hosting the receiver and gathers the links and companion tools, the clock, the language selector and the button that opens the “Information” menu.



Top banner: identification, links, searches, UTC/Local clock, Information menu and languages.

2.1 Station identification line

The first line shows the station name, its location and its Maidenhead locator (Christus Lake - SW France, IN93LR14VR). It contains two buttons:

- **ORSAT** — opens the station sysop’s QRZ.com page.
- **Christus Lake - SW France, IN93LR14VR** — opens a satellite map centred on the receiver’s QTH.

2.2 Second line: contact, users and statistics

- **e-mail** — opens a message pre-addressed to the sysop (question, problem report…).
- **Users** — window listing the users connected in real time (see below).
- **Stats** — dashboard of connection statistics (see below).
- **Other servers: 1, 2, 3** — directories of other WebSDRs: 1 = sdr-list.xyz, 2 = sdr.shbrg.nl, 3 = websdr.org.



“Connected users” window: location, frequency, mode, connection time and duration, map.

The **Connected users** window lists each listener (approximate location, frequency and mode, connection time and duration). Your own line is marked **YOU**. The **TUNE** button on a line tunes

your receiver to the same frequency and mode as that user. A map shows where listeners are; the *Polling* indicator confirms automatic refresh.



“User statistics” window: period, counters and hourly distribution.

The **Statistics** window shows the receiver’s activity over a chosen period (start and end dates, or the **Today**, **7 days**, **30 days**, **All** buttons, then **Load**): number of events and sessions, unique addresses, number of locations, total airtime, average and longest session, and a histogram of the hourly distribution (UTC). Both windows close with the cross or the Esc key.

2.3 Third line: frequency and callsign search

- **MW** / **SW** — databases to identify the broadcast station you are hearing on medium or short wave.
- **PSK Reporter** — community map of digital-mode reception reports.
- **WSPRnet** — database of WSPR beacon reports.
- **Callsign search** — type a callsign in the field, then click **Search**.

2.4 UTC / Local clock

On the left, a digital clock shows the time and date. The **UTC** button displays Coordinated Universal Time — the reference for digital modes, broadcast schedules and the DX cluster — and the **LOCAL** button your computer’s local time.

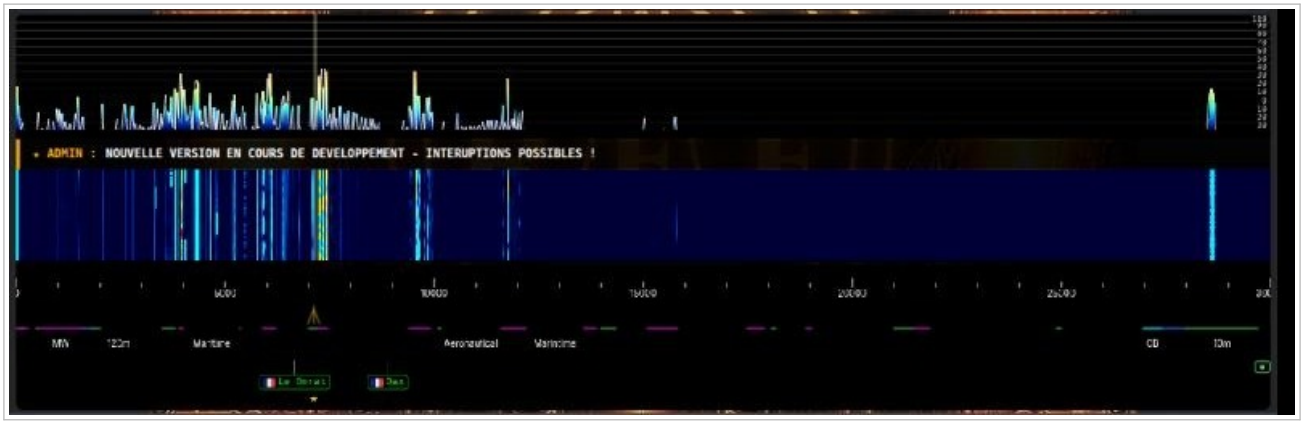
2.5 Language and “Information” menu

On the right are the language buttons described in section 1.2. In the centre, the **Open Meteo - Alerts - Infos - Propagation - DXcluster and User Manuals** button unfolds the “Information” menu detailed in section 13; it then becomes **Close windows** to fold it again.

3. Main screen: spectrum and waterfall

The central area shows two complementary views of the received signal:

- **The spectrum** (upper part) — signal amplitude versus frequency, right now.
- **The waterfall** (scrolling part) — spectrum history over time; colour shows the received intensity.
- **The frequency scale** and, below it, the **band and service markers** (MW, 120m, Maritime, Aeronautical, CB, 10m…).



Spectrum, administrator message, waterfall, frequency scale, band markers and listener labels.

3.1 Tuning with the mouse and wheel

- **Click** on the spectrum or waterfall — tunes the receiver to the clicked frequency; dragging moves the filter.
- **Horizontal drag** — scrolls the visible part of the band without changing the listening frequency.
- **Wheel** — zoom in / out.
- **Shift + wheel** — tune in 0.1 kHz steps; **Ctrl + wheel** — 0.01 kHz steps.
- **Shift + Ctrl + wheel** — snaps the frequency to the round kilohertz (.00 kHz).

3.2 Listener labels and administrator message

Below the scale, small flag labels (e.g. *Le Dorat*, *Dax*) show the frequency each other connected user is listening to; your own position is marked with a star. The small round ● button on the right of the area **hides or shows** these labels.

When a ★ **ADMIN** banner runs across the waterfall, it is a message broadcast by the sysop to all listeners (maintenance, new version, planned interruption…).

> ***Tip:*** *Waterfall colours, brightness and contrast are set in the right column (section 10).*

4. Dial, bands, modes and filters

The middle column gathers everything related to tuning the receiver.



Middle column: dial and S-meter, diversity, fine tuning, VFO / Modes / Bands / IF Filters / TCI-CAT, scanner and decoders.

4.0 Frequency dial, S-meter and indicators



Dial: UTC and local time, frequency, band, VFO, mode, filter width, status indicators and S-meter (dark style).

The dial shows UTC and local time, the tuned frequency (e.g. *007,120.000* kHz), then the current band, the active VFO, the mode and the filter width (*Current Band: 40m — VFO A | LSB | 2.70 kHz*).

- **Direct tuning on the digits** — hover a digit of the frequency and turn the wheel to change it; a **right-click** on the dial lets you type a frequency directly.
- **Status indicators** — **MUTED**, **SQ** (squelch), **AI** (AI filter), **NR**, **NB**, **NS**, **AN** and **CTCSS** light up when the function is active: one glance tells you which processing is running.
- **Analogue S-meter** — scaled from S1 to +60 dB, with four readings under the needle: **dBm** and **dBµV** (received level), **SNR** (signal-to-noise ratio) and **NF** (noise floor).

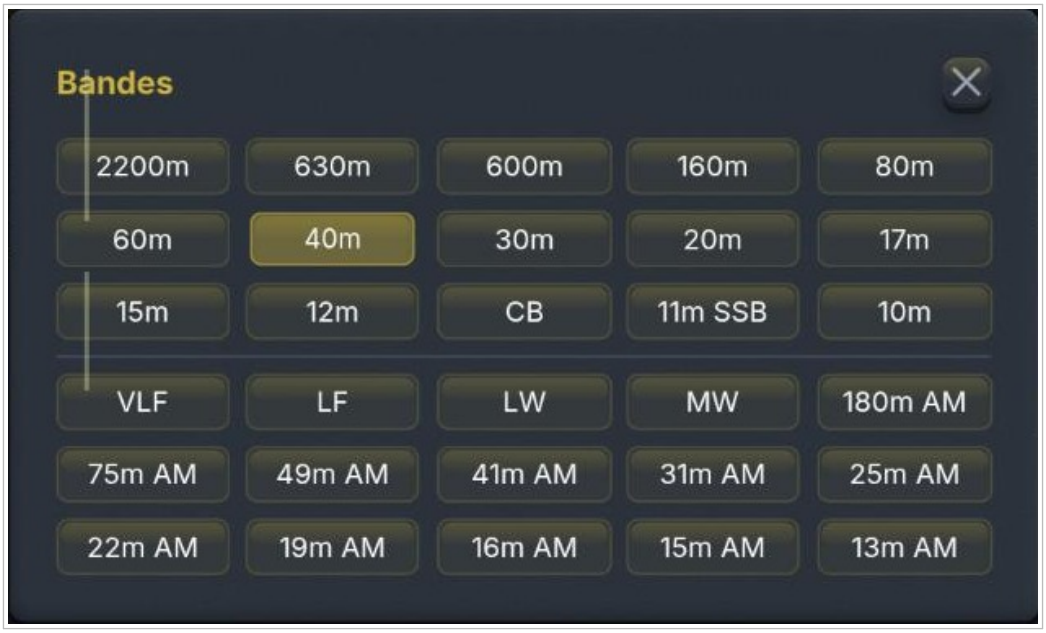
A **click on the S-meter** cycles through its three styles: **dark**, **light** and **vintage**.



“Light” (left) and “vintage” (right) S-meter styles.

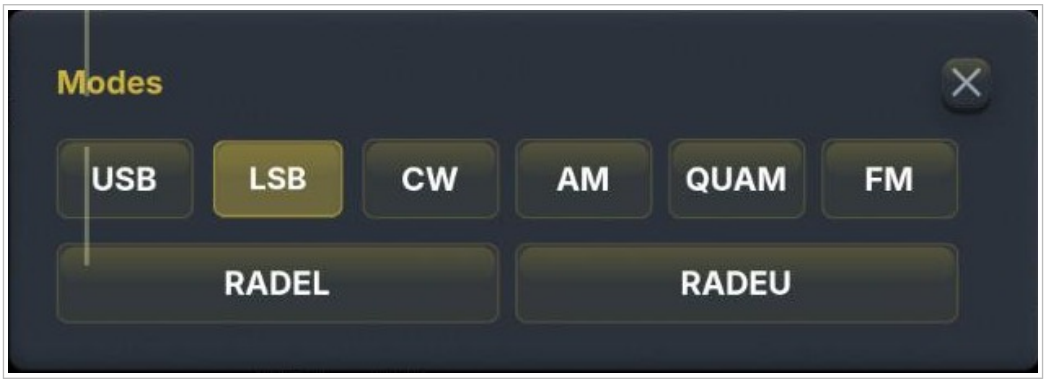
4.1 Band selector

The **Bands** button opens the list of amateur bands (2200 m to 10 m, CB, 11 m SSB) and broadcast bands (VLF, LF, LW, MW, 180 m to 13 m AM). One click tunes the receiver to a representative frequency of the band, in the usual mode; the current band is highlighted.



Band selection window.

4.2 Demodulation modes

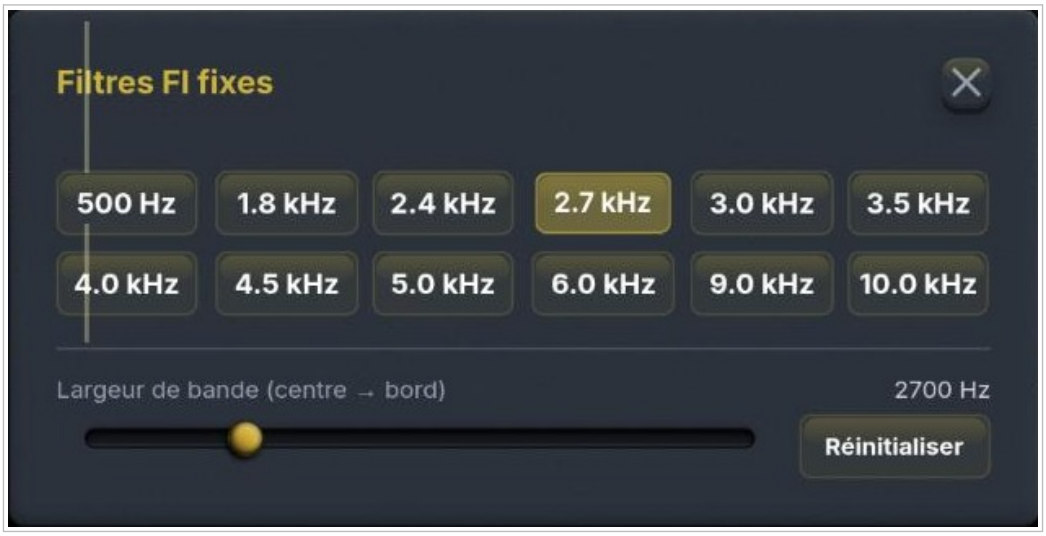


Demodulation mode selection window.

- **USB / LSB** — single sideband (by convention LSB below 10 MHz, USB above).
- **CW** — telegraphy, with a narrow filter suited to Morse.
- **AM** — classic amplitude modulation; **QUAM** — quadrature AM demodulation.
- **FM** — frequency modulation.
- **RADEL / RADEU** — decoding of **FreeDV** digital voice (RADE mode) on the lower / upper sideband.

4.3 IF filters

The **IF Filters** button opens the “Static IF Filters” window: twelve preset widths from 500 Hz to 10 kHz, and a **Bandwidth (centre → edge)** slider for continuous adjustment, with the value in Hz. The **Reset** button returns to the mode’s default width.



IF filter window.

> ***Tip:*** *A narrow filter cuts noise and adjacent interference but may clip part of the wanted signal. In CW, a few hundred hertz greatly improves copy of weak signals.*

4.4 VFO A / B

The **VFO A** button switches between two VFOs: keep one frequency on one VFO and explore the band with the other, then return to the first with one click.

4.5 TCI-CAT

The **TCI-CAT** button opens the “CAT control via TCI” window, which synchronises the web interface with software running on your computer (**ExpertSDR**, **AetherSDR**, **Thetis**): the usual TCI ports 50001 and 40001 are probed automatically.



“CAT control via TCI” window.

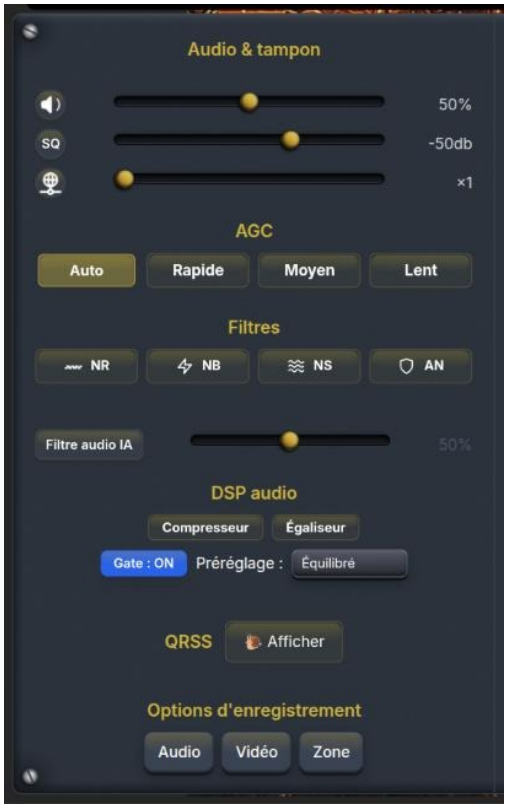
- **Status** — *Inactive*, *Wait* or connected; the dot in front of **TCI-CAT** on the button reflects it.
- **CAT Sync** — enables frequency synchronisation: a change on one side is mirrored on the other.
- **Host** — machine running the TCI software (*localhost* by default, or its IP address on the local network).

***Tip:** This link connects the software on your computer to the tuning done in the browser; it does not key the site’s transmitter and does not replace a direct CAT link to a transceiver.*

4.6 Fine tuning

The **Fine Tuning (kHz)** row shifts the frequency by fixed steps: **-5, -1, -0.5, -0.1, -0.01, 0, +0.01, +0.1, +0.5, +1, +5** kHz. Small steps centre a CW or digital signal, large ones jump from one signal to the next.

5. Audio control column (left)



Left column: Audio & Buffer, AGC, Filters, AI Audio Filter, Audio DSP, QRSS and Recording Options.

The left column gathers the sound output and processing settings.

5.1 Audio & Buffer

- **Volume** (speaker) — listening level; clicking the icon mutes / unmutes (MUTED indicator).
- **SQ** (squelch) — the slider sets the threshold in dB. The **SQ** button offers an **automatic squelch**: go to a quiet frequency, click to measure the noise and the threshold sets itself; click again to disable it.
- **Buffer** (globe icon) — audio buffer multiplier (×1 by default). Increase it on a slow or unstable Internet connection, at the cost of a little latency.

5.2 AGC

Automatic gain control keeps a steady listening level. **Auto** (recommended) picks a suitable speed; **Fast** suits CW and impulsive signals; **Medium** most SSB use; **Slow** AM and broadcast. Some decoders (QRSS, for example) force their own AGC, shown next to the title.

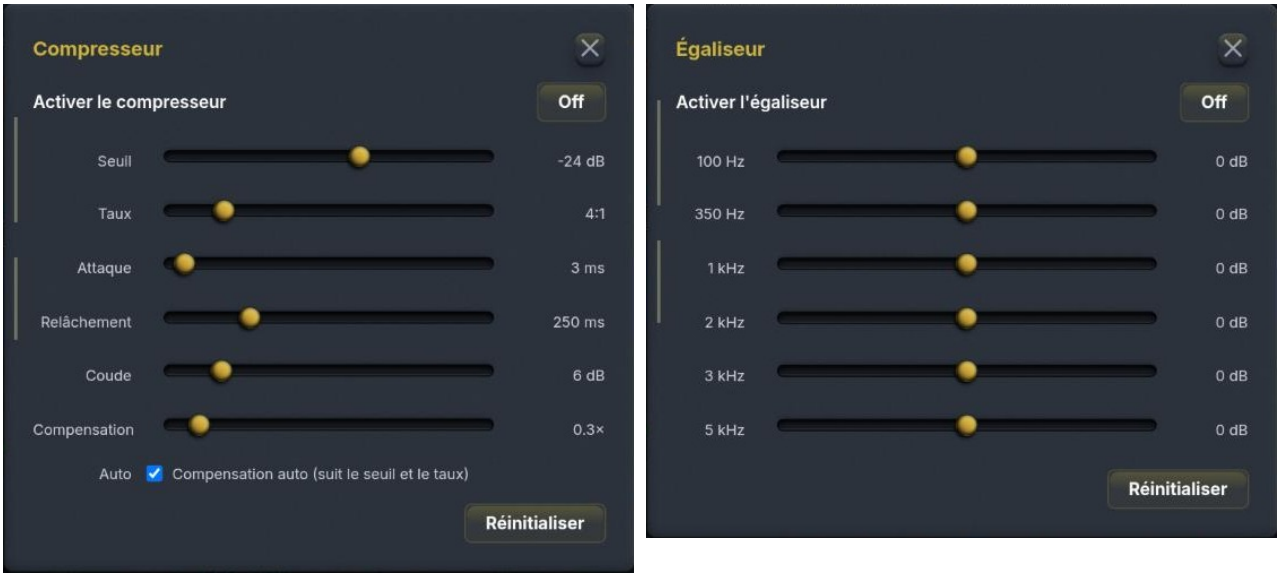
5.3 Noise filters and AI audio filter

- **NR** — wideband noise reduction (continuous background noise).
- **NB** — impulse noise blanker (ignition noise, clicks).
- **NS** — background hiss suppression.
- **AN** — automatic notch: detects and attenuates steady carriers (heterodynes).
- **AI Audio Filter** — neural noise reducer, active in **SSB and AM/SAM only**. The slider sets its strength: up to 50 %, the original gentle setting; above, strong filtering (up to 40 dB of reduction, silences cleaned). Turn it down if QRP stations become inaudible.

> ***Tip:*** *Enable filters one at a time while listening: stacking several aggressive processes often hurts intelligibility instead of helping it.*

5.4 Audio DSP: compressor, equalizer and gate

The **Compressor** and **Equalizer** buttons each open a settings window with its own enable switch and a **Reset** button.



Compressor (left) and six-band Equalizer (right) windows.

- **Compressor** — **Threshold** (dB where compression starts), **Ratio** (e.g. 4:1), **Attack** and **Release** (ms), **Knee** (transition softness, dB) and **Makeup** gain; the **Auto makeup** box computes that gain from threshold and ratio.
- **Equalizer** — six bands adjustable in gain: 100 Hz, 350 Hz, 1 kHz, 2 kHz, 3 kHz and 5 kHz.
- **Gate: ON / OFF** — noise gate running **on the server**, engaged by default; it silences the noise between words or transmissions.
- Gate **Preset**: *Balanced*, *Aggressive*, *Weak Signal*, *Smooth*, *Maximum*, *CW/Digital* and *AM/FM*.

Tip: *To lift a voice out of the noise, slightly cut 100–350 Hz and slightly boost 1–2 kHz, where speech intelligibility is concentrated.*

6. QRSS

The **Show** button opens the QRSS settings and adds, below the waterfall, a very high resolution spectrogram for extremely slow, very low power signals (Morse at several seconds per dot, DFCW). The button becomes **Hide** to close it all.



- **Band** — list of known QRSS windows (10140.00 kHz on 30 m — the main one —, 7039.90 and 7000.85 kHz on 40 m, 14096.90 kHz on 20 m, 80 m, 160 m, 630 m, 2200 m, 60 m, 17 m…), then **Tune**.
- **Speed** — QRSS 3, 10, 30…: sets the fineness of the time-frequency analysis.
- **Centre** — observed audio centre frequency (Hz).
- **Span** — window shown around the centre; **Clear** resets the display.
- **Gain** — overlay contrast (dB).
- **Colour** — overlay palette (Rainbow…).
- **Save as .PNG** — saves an image of the overlay.

QRSS settings panel.



QRSS overlay: audio scale in Hz on the left, resolution indicators at the bottom (speed, Hz/bin, window length, s/column, Hz/row).

The indicator line under the overlay recalls the current analysis resolution. When the vertical resolution is flagged as the display limit, reduce the **Span** to gain definition. While QRSS is on, the AGC automatically switches to slow.

> ***Tip:*** *QRSS traces build up over several minutes: let the overlay run on the chosen window (for example 10 140 kHz on 30 m) before concluding there is no signal.*

7. Recording

The **Recording Options** section offers three local captures, saved on your computer:

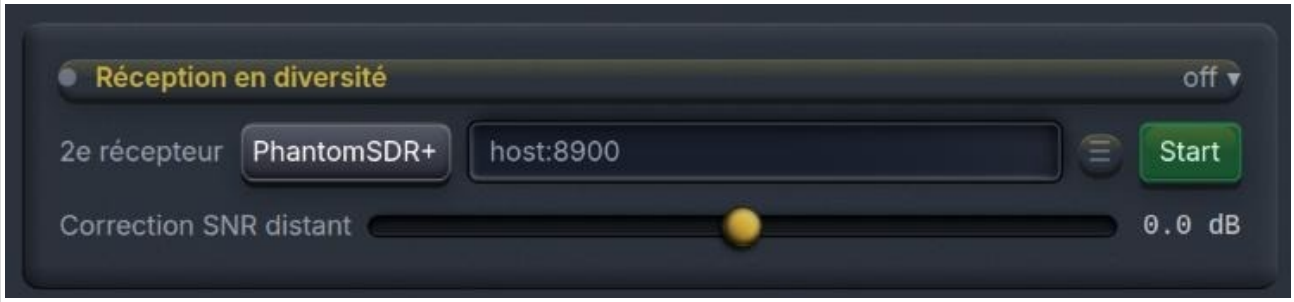
- **Audio** — records the demodulated audio only, as WAV.
- **Video** — records the waterfall together with the demodulated audio.
- **Area** — draw an area on the waterfall with the mouse to record only that part; with no area set, the whole panel is recorded.

A second click stops recording and offers the resulting file for download.

> ***Tip:*** *The recording captures exactly what you hear, filters and DSP included: check mode, frequency and settings before starting.*

8. Receive diversity

Receive Diversity combines the ORSAT receiver with a **second, remote receiver** reachable over the Internet, to ride through fading and improve the signal-to-noise ratio thanks to two geographically different listening points. Click the bar to unfold the panel.



“Receive Diversity” panel unfolded.

- **Switch** (off / on, top right) — enables or disables the combination.
- **2nd receiver** — type of remote receiver: **PhantomSDR+**, **KiwiSDR**, **UberSDR** or **WebSDR**.
- **Address** — host and port of the remote receiver (*host:port*).
- ☰ **Saved receivers** — stores your favourite receivers by type: **Add** one by hand, then recall it with a click; **import** reloads a list from a backup file.
- **Start** — connects to the remote receiver and starts combining.
- **Remote SNR trim** — compensates (in dB) the level difference between the two receivers to balance their contributions.

> ***Tip:*** *The remote receiver must cover the same frequency and be reachable; results depend on its reception quality and on network latency.*

9. Scanner and decoders



Fine tuning, VFO / Modes / Bands / IF Filters / TCI-CAT, scanner and decoders control.

9.1 Scanner

- ‹ / › — scan down or up in 1 kHz steps; scanning resumes after each stop.
- ■ — stops scanning.
- ◐ — excludes the current frequency from the scan (carrier or permanent interference).
- **Threshold (e.g. 30 dB ▾)** — opens the scanner settings menu (below).



- **Range** — *Whole band* or only the *Visible part* of the waterfall.
- **Sweep** — *Every channel* or *Skip empty*.
- **Stop at** — level above the noise floor (+6 to +40 dB) that stops the scan on the first active channel.
- **Resume after** — *Hold* (stays on the signal), 3 s, 5 s or 10 s after it disappears.
- **Max dwell** — maximum time on one signal: Off, 30 s, 1 min or 2 min.

Scanner settings menu.

9.2 Digital decoders

The **Decoders Control** row offers eleven decoders, each started with one click (a second click stops it):

Button	Decoder	Use
FT8 / FT4 / FT2	WSJT weak-signal digital modes	DX traffic, propagation
JS8	JS8Call	Weak-signal keyboard messaging
CW	Morse telegraphy	Automatic Morse copy
WSPR	WSPR beacons	Propagation study
FAX	HF FAX / WEFAX	Radio-facsimile weather charts
SSTV	Slow-scan television	Amateur pictures
NAVTX	NAVTEX	Navigational warnings (518 / 490 kHz)
HFDL	HF Data Link	Aircraft ACARS messages on shortwave
FLDIGI	fldigi family	RTTY, SITOR, PSK31, Olivia, MFSK, DominoEX, THOR, THROB, Hellschreiber, MT63, Packet, APRS



FT8 decoder: audio waveform with decode markers, sync offset and message table.

- **Audio waveform** — yellow markers flag each decoded signal in the passband.
- **Farthest Distance** — distance of the farthest contact since the session started.
- **Sync offset** — timing correction, adjusted automatically when **Auto** is ticked.
- **Table** — one row per decode: MODE, DB (level), HZ (audio frequency), DT (timing offset), MESSAGE, LOCATOR (Maidenhead grid) and DIST (estimated distance).

> ***Tip:*** *The FT8/FT4/WSPR decoders require tuning to the mode’s frequency (e.g. 7074 kHz USB for FT8 on 40 m).*

9.3 Decoder ID: automatic mode identification

The **Decoder ID** switch (Off / On) continuously analyses the signal in the passband and proposes the recognised digital mode, with a confidence percentage and its measured characteristics (width, shift, baud rate, burst length, S/N). Alternative modes may be suggested.



Decoder ID on: FT8 recognised at 81 %, “Re-centre & retry” and “Use FT8” buttons.

- **Use …** — starts the decoder matching the recognised mode with one click.
- **Re-centre & retry** — re-centres the signal in the filter and restarts the analysis.

- □ — measurement details.

> ***Tip:*** *Decoder ID is off by default: it only uses CPU while it is running.*

9.4 Identify a signal (Artemis database)

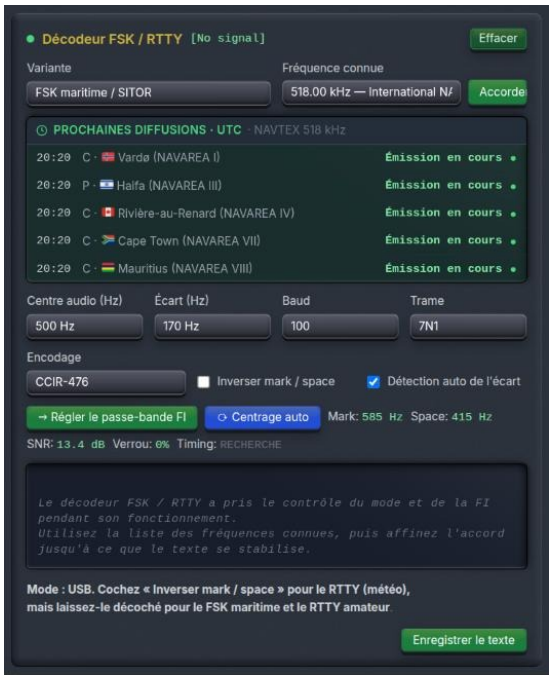
The **Identify** button searches a reference signal database (the Artemis database) for what matches the current frequency, filter width and demodulation. The window ranks known signals by match percentage; a **Filter** field searches by name, category or modulation, and the Frequency / Width / Demodulation fields can be edited.



“Identify a signal” window: ranked list and detailed sheet (spectrogram, audio sample, characteristics, description).

Clicking an entry shows its **sheet**: reference spectrogram, an **audio sample** to compare by ear, width, modulation, reception mode, frequency range, category, area and description.

9.5 FLDIGI decoder (FSK / RTTY and other modes)



- **Variant** — maritime FSK / SITOR, weather RTTY, amateur RTTY, PSK31, Olivia, MFSK16·32·64, DominoEX, THOR, THROB, Hellschreiber, MT63, Packet (AX.25), APRS.
- **Known frequency** — list of reference frequencies (NAVTEX 518 / 490 kHz, HF SITOR MSI, NBDP…), then **Tune**.
- **Upcoming broadcasts · UTC** — schedule of NAVTEX stations on air and coming up.
- **Audio centre**, **Shift**, **Baud**, **Framing** and **Encoding** (CCIR-476, ITA2/Baudot, ASCII).
- **Invert mark / space** — tick it for weather RTTY, leave it unticked for maritime FSK and amateur RTTY; **Auto shift detection**.
- **Set IF passband** and **Auto centre**; Mark / Space, SNR, Lock and Timing indicators.
- Decoded text area, **Clear** and **Save text**.

FLDIGI decoder, maritime FSK / SITOR variant (NAVTEX 518 kHz).

***Tip:** While running, this decoder takes control of the mode and IF filter: pick a known frequency, then fine-tune until the text stabilises.*

9.6 HFDL decoder



- **Ground station** — choose an HFDL station (Shannon, Reykjavik, Canarias, Riverhead, Johannesburg…) and one of its frequencies, then **Tune & set IF**.
- The **Dial** indicator warns when you are off an HFDL frequency; **Noise**, **Frames**, **Rhythm**, **Messages** and **Squitters** counters follow decoding.
- **Map** — plots decoded aircraft; **Save** and **Clear** the log.
- Tick **Squitters** to also show the ground stations’ beacons.
- Hover a message to see the full frame.

HFDL receiver (dumphfdl decoder in WebAssembly).

***Tip:** Ground stations send a squitter every 32 s and aircraft answer when they have traffic: wait a few minutes and pick a frequency where propagation is open towards the chosen station.*

10. Waterfall controls



These controls, in the right column, only change how the spectrum and waterfall are displayed, never the audio.

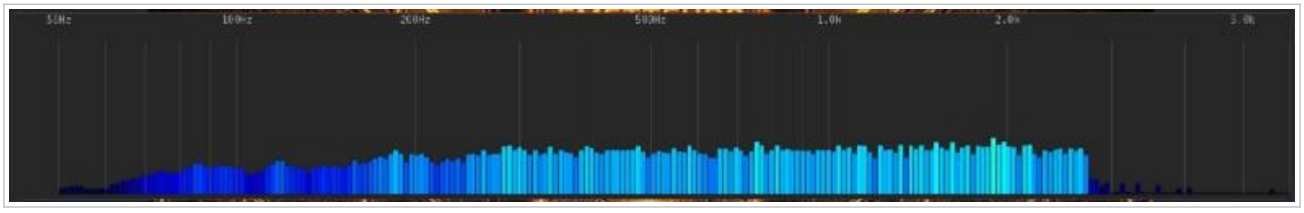
- **Auto Adj.** — automatically calibrates Min / Max from the ambient noise floor.
- **Min** — brightness: lower bound of the displayed intensity (dB).
- **Max** — contrast: upper bound; bringing Min and Max closer makes weak signals stand out.
- **Magnifier and ×1 slider** — enables step-by-step manual zoom and sets its factor.
- **Magic eye** — glowing green bar responding to received signal strength, like on old valve receivers.
- **Zoom** — **To Band** frames the waterfall on the band of the tuned frequency; **In**, **Out**, **Max**, **Min**.
- **Palette** — waterfall colours: turbo, gqrx, twente, twentev2, SpectraVU, jet, custom, PhantomSDR.
- **Connection indicator** — number of connected users and data rate (kbit/s) received by your browser.

Right column: waterfall, zoom, palette, audio spectrogram, bookmarks and connection indicator.

> ***Tip:*** *A good Min / Max setting — or Auto Adj. — makes all the difference between an unreadable waterfall, all black or all white, and a display where weak signals stand out from the noise.*

11. Audio spectrogram

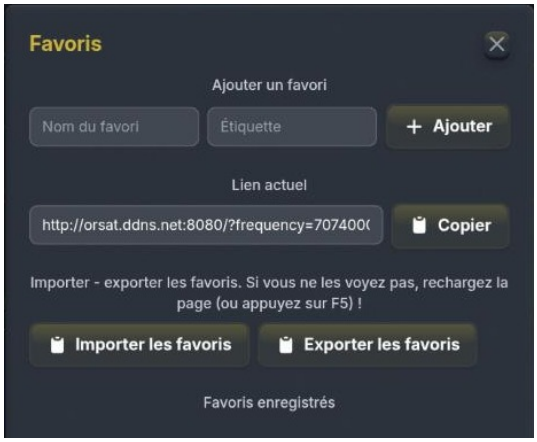
The **Show** button of the “Audio Spectrogram” opens, above the panels, an analyser of the demodulated audio (the station you are hearing) on a logarithmic scale from about 50 Hz to 5 kHz. It becomes **Hide** to close it.



Audio spectrogram: amplitude per audio frequency, logarithmic scale.

- **Gain** — display sensitivity.
- **Colour** — Rainbow, Blue, Green or Greyscale.
- Ideal for centring a digital or CW signal precisely in the filter and for seeing the structure of digital modes.

12. Bookmarks



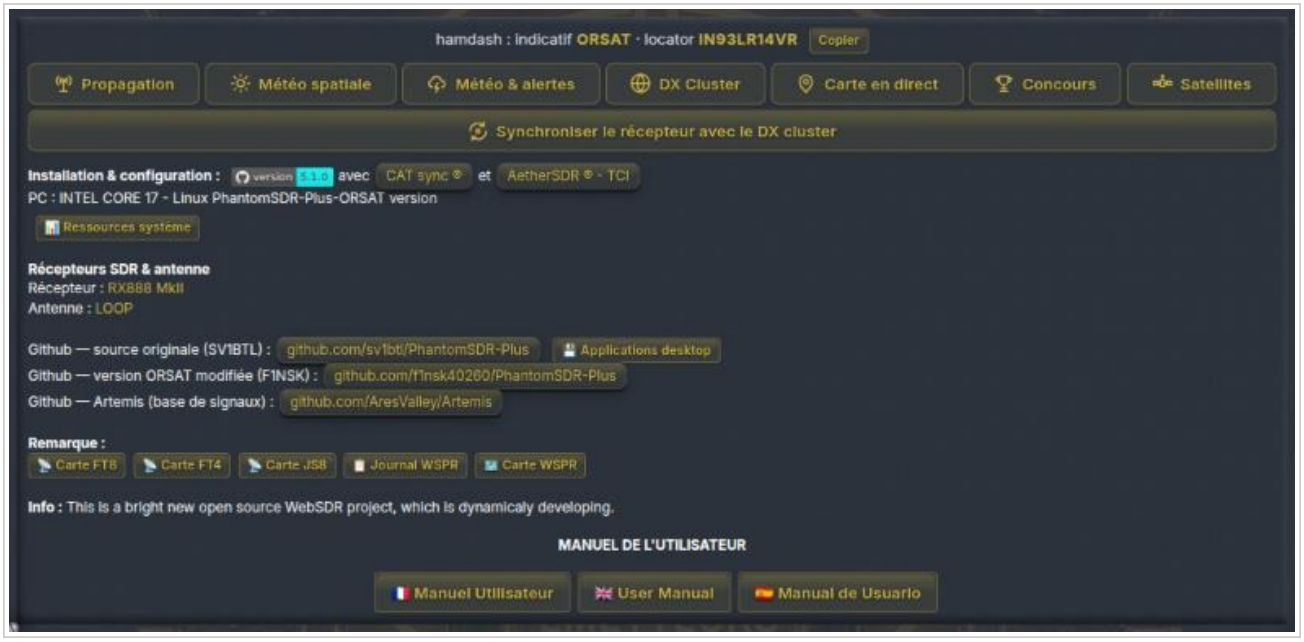
Bookmarks window.

- **Add a bookmark** — enter a *Name* and a short *Label*, then **Add**: the current frequency and tuning settings are stored.
- **Current link** — URL of the current frequency and mode, to share with **Copy**.
- **Import / Export bookmarks** — save or restore the list to or from a file. Reload the page (F5) if imported bookmarks do not appear.
- **Saved bookmarks** — click an entry to tune the receiver.

Tip: *Bookmarks are stored in your browser, on your computer: export them to find them again on another machine.*

13. “Information” menu

The central banner button unfolds this menu, which gathers the companion tools: hamdash, DX cluster synchronisation, station configuration, links and manuals.



“Information” menu unfolded.

13.1 hamdash tools

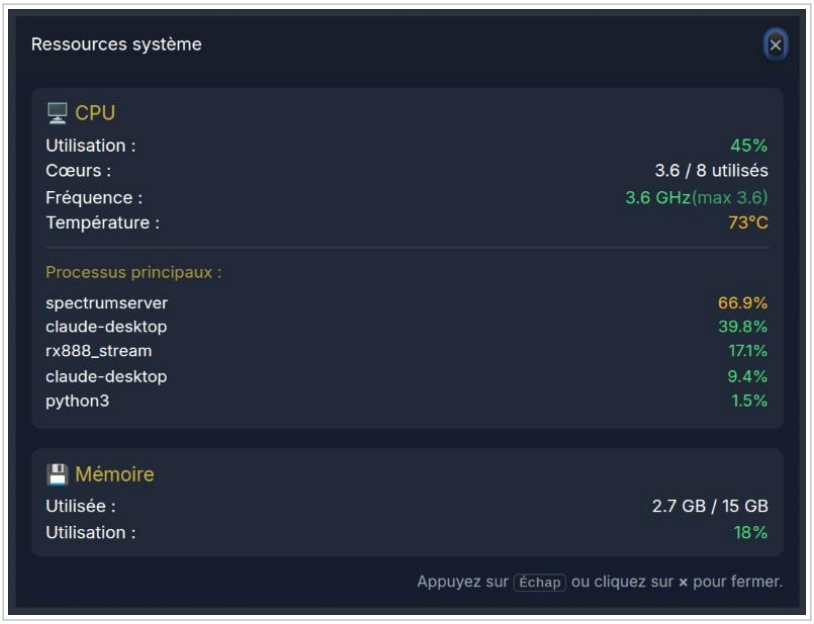
The **hamdash** line recalls the station callsign (ORSAT) and locator; **Copy** puts them on the clipboard to paste into hamdash. The seven buttons open the matching dashboards:

Button	Content
Propagation	HF propagation forecasts
Space weather	Solar wind, K and A indices, flares
Weather & alerts	Local weather and thunderstorm alerts
DX Cluster	Live DX cluster spots
Live map	Map of active bands and stations
Contests	Contest calendar
Satellites	Amateur satellite passes

The large **Sync the receiver with the DX cluster** button opens the window described in section 14.

13.2 Setup & configuration

- **version 5.1.0** — PhantomSDR-Plus version, linked to the GitHub repository.
- **CAT sync** and **AetherSDR - TCI** — synchronisation features available on this receiver (section 4.5).
- **PC** — server hardware and operating system.
- **System resources** — server load window (below).



“System resources” window: processor, top processes and memory.

It shows processor usage and frequency, the number of cores in use, its temperature, the most demanding processes and memory use — useful to understand a slowdown. Close with Esc or the cross.

13.3 Receiver, antenna and links

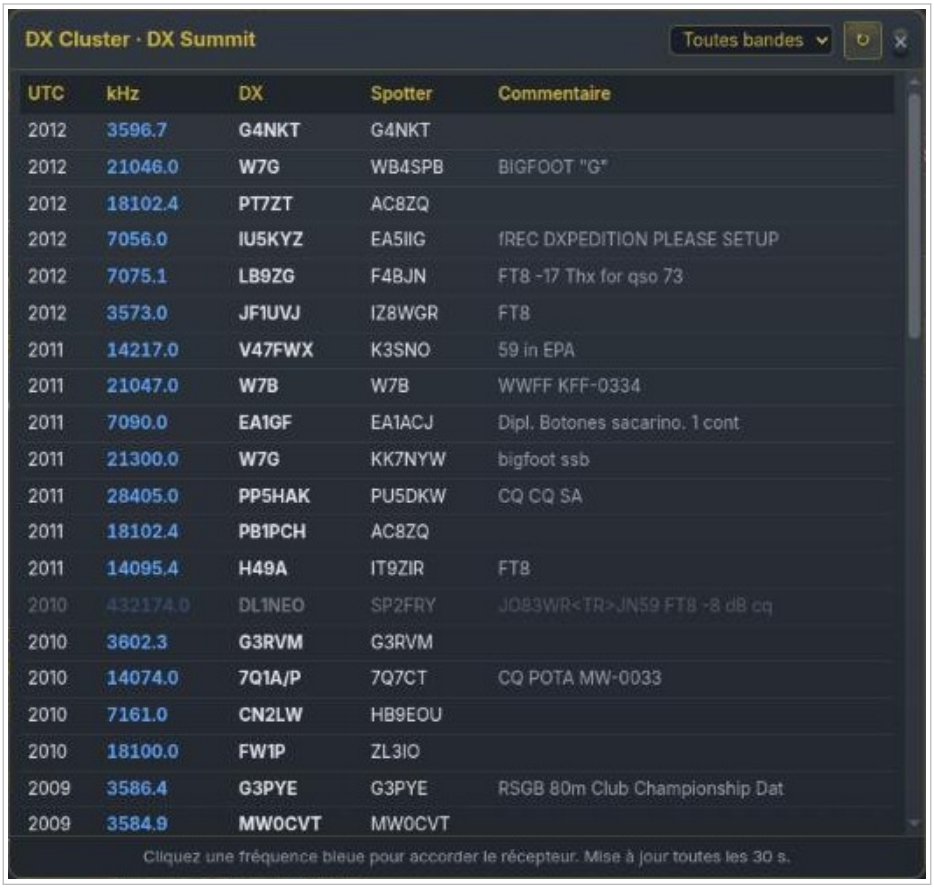
- **Receiver**: RX888 MkII — **Antenna**: LOOP.
- **Github — original source (SV1BTL)**, **ORSAT modified version (F1NSK)** and **Artemis** (signal identification database).
- **Desktop applications** — download PhantomSDR+ Desktop for Linux and Windows, to listen from a native application.
- **Note** — quick links to the PSK Reporter **FT8**, **FT4**, **JS8** maps, the **WSPR log** and the **WSPR map**.

13.4 User manuals

The **User manual** section gathers the three versions of this document: **Manuel Utilisateur** (French), **User Manual** (English) and **Manual de Usuario** (Spanish). Each opens in its own window, closed with the cross or the Esc key.

14. DX Cluster

The **DX Cluster · DX Summit** window shows, in real time, the spots posted by the worldwide amateur radio community: UTC time, frequency (kHz), DX callsign, spotter and comment.



DX Cluster window: blue frequencies are clickable, grey ones are outside the receiver’s coverage.

- **Click a blue frequency** — immediately tunes the receiver to the spot.
- Greyed spots are on frequencies the receiver does not cover (VHF/UHF, for example).
- **Band filter** — All bands, or one band from 160 m to 6 m.
- **↻** — immediate refresh; the list also updates itself every 30 s. **×** closes the window.

> ***Tip:*** *The DX cluster and the Users window (TUNE button) are the two quickest ways to “follow the DX” spotted by other operators or heard by other listeners.*

15. Tips, troubleshooting and good practice

15.1 Listening quality

- First choose an IF filter suited to the mode (wide for AM/FM, narrow for CW), then enable noise filters one at a time.
- In SSB or AM, try the **AI Audio Filter** at 50 % before pushing its strength.
- Set Min / Max (or Auto Adj.) to clearly separate weak signals from the noise floor.

15.2 Audio dropouts or latency

- Increase the audio **buffer** (section 5.1) on a slow or unstable connection.
- Check **System resources** (section 13.2): a high server load can explain a slowdown.
- Watch the **data rate** shown at the bottom of the right column.

15.3 Decoders

- When unsure about a signal, turn on **Decoder ID** or use **Identify** before choosing a decoder.
- Run only one decoder at a time and tune to the mode’s own frequency.

15.4 Shared use

- Several listeners share the same receiver, each with their own tuning and audio settings, without affecting the others.
- The labels below the waterfall and the Users window show where the others are listening; the ★ ADMIN message keeps you informed about the site.

*End of the user manual — English version. For any question, use the **e-mail** link in the top banner to contact the station sysop.*